What Does It Really Cost to Get Sick?

Exercises on the Kibbo Healthcare Cost Dataset (2026), reduced edition: 58 rows, 4 countries (United States, United Kingdom, Germany, Spain), 4 treatments

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Teacher edition. This version includes the answer key at the end. Do not hand it to students; a separate student sheet without answers is available.

Level: university (the critical-thinking questions also work for upper high school). **Subjects:** health economics, public policy, statistics, economics. **Tools:** Excel or Google Sheets. No programming needed.

Before you start

The dataset splits cost into layers: **Layer A** is what you pay every year whether you get sick or not (premiums, contributions, taxes); **Layer B** is what a specific treatment costs when you get it; **Layer C** is optional private insurance, shown separately. Read the README first. This is the **reduced edition** of the dataset: every row is still in the file, but the value of 32 rows is withheld because the source's own terms do not allow Kibbo to republish it, and the *value_status* column says what happened to each row. The empty cells are part of what you are going to analyse.

Files used (download links and the README are on the web page of this sheet: getkibbo.com/education/exercises/healthcare-costs):

- kibbo-healthcare-cost-dataset-2026.csv
- healthcare-dataset-README.md

Exercises

Exercise 1 — Reading the data *(basic)*

1. Use the Layer A rows for Germany and Spain.
   a) Add Germany's general contribution rate and the average additional contribution. What is the total rate?
   b) The notes say both parts of the German rate are split equally between employer and employee. What percentage of salary does the German employee pay? (Ignore contribution ceilings.)
   c) What percentage does the Spanish worker pay, according to the dataset?

Exercise 2 — Reading the status column *(basic)*

1. Use the value_status column (filter or pivot table). How many rows are there in each group: "kept", "data gap", "not applicable", and withheld (every other status)? How many cells in the value column are empty?

Exercise 3 — Comparing the wrong things *(advanced)*

1. Spain's total contribution rate is 28.3%. Germany's statutory health insurance contribution is 14.6% plus an average 2.9% additional contribution. A student concludes: "Spanish workers pay almost twice as much for healthcare as German workers." Use the notes column to explain why this conclusion is wrong.

Exercise 4 — Wait times that can't be compared *(advanced)*

1. The wait-time rows show: Germany 36 days (time to a specialist appointment), the United Kingdom 11.9 weeks, about 83 days (time to treatment), Spain 121 days (time to surgery) and Spain 102 days (time to first specialist consultation). The US wait-time row exists, but its value is withheld.
    a) Why is it wrong to conclude from these numbers that Germany has the shortest wait and Spain the longest?
    b) Which rows can be compared with each other, and what do they show?
    c) A student adds Spain's 102 days and 121 days and says a patient waits 223 days from first consultation to surgery. What is wrong with this?

Exercise 5 — Two numbers for one question *(intermediate)*

1. The notes on the German wait-time row say that the same official answer also gives 42 days (up from 33 in 2019) for the patients who waited at least one day, and that the dataset uses 36 days instead.
    a) What does each figure (36 and 42) measure? Why is 42 higher?
    b) A journalist writes: "Waiting times in Germany rose from 33 days in 2019 to 36 days in 2024." What is wrong with this sentence?

Exercise 6 — Two kinds of empty cell *(advanced)*

1. Compare the 11 "data gap" rows with the withheld rows.
    a) What is the difference between the two kinds of empty cell? Which countries and which layer have the data gaps?
    b) Why does the dataset leave data gaps empty instead of using an estimate?
    c) What can a reader still do with a withheld row, and what can't they do?
    d) What does the pattern of empty cells say about the transparency of health-cost information? Give at least two different ways in which information can be "not available".

Exercise 7 — Following a figure to its source *(intermediate)*

1. Take the US Layer A row "Worker's annual premium share (employer-sponsored single coverage)". Its value is empty in this edition.
    a) Open the link in the source_url column. In the report, find the average annual worker contribution for single coverage and the average annual deductible for single coverage. Note the page or section where you found each.
    b) Add the two figures. What does the total represent, and what does it leave out?
    c) Write a one-line citation for each figure (author, title, year, link).

Exercise 8 — Currency and time *(advanced)*

1. The UK hip replacement row shows £6,061 and is labelled “historical reference only”. Give two reasons why you shouldn’t compare it directly with US hospital prices such as those in the sources linked from the US Layer B rows.

Connect it to real life

What can you do if you think a medical bill is wrong? Use the Kibbo Healthcare Billing Dispute & Refund Letter Generator and the Medical Billing Dispute / Insurance Appeal Kit (getkibbo.com).

Answer key

Exercise 1

a) 14.6% + 2.9% = 17.5%.

b) 17.5% ÷ 2 = 8.75% (7.3% + 1.45%).

c) 4.7%. Do not stop here: the two numbers are not directly comparable, which is the point of Exercise 3.

Exercise 2

1. Kept: 11. Data gap: 11. Not applicable: 4. Withheld: 32 (iFHP/HCCI 16, MyCareCost 7, KFF 5, BMJ / The King’s Fund 3, Merritt Hawkins 1). Total: 58.

   Empty value cells: 43 = 11 data gaps + 32 withheld. The 4 “not applicable” rows are not empty: their value is 0 (emergency cases, no wait).

Exercise 3

1. Spain’s 28.3% is not health-specific: it also covers sick leave, maternity/paternity and other benefits. Most of it (23.6 points) is paid by the employer; the worker’s share is 4.7%. Germany’s rate is health insurance only and is split equally between employer and employee. The two numbers measure different things.

Exercise 4

a) They measure different points in the care pathway. Germany’s figure is time to a specialist appointment; the UK figure and Spain’s 121 days are time to treatment. The UK figure is also a median across all specialties, not hip-specific.

b) Germany (36 days) and Spain (102 days) both measure time to a specialist appointment: Spain’s is about 2.8 times Germany’s (the years differ, 2024 and the second half of 2025, so treat it as approximate). The UK (about 83 days) and Spain (121 days) both measure time to treatment, with the median/average caveat above.

c) The two figures come from different measurements and are not consecutive stages for the same patients. Nothing in the dataset says they describe the same people, so adding them does not give one patient’s journey.

Exercise 5

a) 36 days is the 2024 average for all GKV-insured patients, including those seen the same day. 42 days only counts patients who waited at least one day. Leaving out the zero-day waits necessarily raises the average.

b) The 33 days in 2019 belongs to the “waited at least one day” group, so it can only be compared with 42 days: 33 to 42 is a rise of about 27%, not the 9% suggested by 33 to 36. The dataset has no 2019 figure for all patients.

Exercise 6

a) A data gap is a figure that was not found in any source: the public-system internal cost per treatment. It affects the UK (3 rows: appendectomy, knee MRI, vaginal delivery), Germany (4) and Spain (4), all in Layer B. A withheld value exists, but its publisher’s terms do not allow Kibbo to republish it.

b) An estimate would present a guess as data. The gap is itself a finding.

c) They can follow the source and source_url columns to the publisher’s own page and read the figure there, under the publisher’s terms. They cannot rebuild any comparison that depends on that value from this file alone.

d) Strong answers separate at least two things: whether the information exists in published, comparable form (the public-system costs per treatment), and whether it can be freely reused (the withheld figures). Information can be missing, restricted by licence or terms, or gated behind membership.

Exercise 7

a) The figures are in the report the link points to (the 2025 employer health benefits survey); check each student’s numbers and page references against it. The values are deliberately not printed here: they stay with their publisher.

b) It is roughly what a worker with single coverage pays in a year through the premium contribution and the deductible, before most insurance cost-sharing applies. It leaves out coinsurance (the percentage paid after the deductible), copays, costs above the deductible up to the plan’s out-of-pocket maximum, and the share of the premium paid by the employer.

c) Accept any consistent format that names the organisation, the report title and year, and the link.

Exercise 8

1. Strong answers mention any two of: it is in pounds, not dollars; it is from 2017/18, so it doesn’t reflect inflation since then; it is a public-system internal cost, a different kind of figure from a price billed to a patient or insurer.